The AI-driven Decision-Making (AIDM) Framework: Integrating AHP and ChatGPT-4 for Supplier Selection

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Abstract

The integration of AI into decision-making is transforming industry norms by enabling smarter, faster, and more reliable decisions. This paper presents a novel approach for supplier selection in manufacturing by combining the Analytic Hierarchy Process (AHP) with the Generative Pre-trained Transformer 4 (GPT-4). Using GPT-4 as virtual agents to mimic expert evaluations automates the supplier selection process, significantly improving efficiency. The study demonstrates the effectiveness of large language models as virtual experts in complex decision-making, highlighting AI's crucial role in strategic manufacturing operations. Comparing GPT-4's assessments with human expert judgments confirms the model's reliability and effectiveness in optimizing supplier selection. This approach enhances decision-making processes and emphasizes the broad utility of AI-centric methodologies in various industries.

keywords: Analytic Hierarchy Process (AHP); Large Language Models (LLMs); AI-enabled Decision Making, Supplier Selection, Generative AI

1 Introduction

The advancement of Large Language Models (LLMs), such as OpenAI's ChatGPT (Generative Pre-trained Transformer), has had significant impacts across various domains. The widespread popularity and acceptance of ChatGPT are remarkable, attracting millions of users daily (1). Trained on massive and diverse datasets from the internet, ChatGPT includes extensive information making its knowledge base comprehensive enough to parallel subject domain experts in many fields. Recent studies show the effectiveness of ChatGPT in medical decision-making (2), supplier selection (3), and business decision-making, such as drafting marketing content and brainstorming ideas (4).

Due to this potential, the integration of Generative AI (GenAI) can revolutionize the decision-making domain, especially in Multi-Criteria Decision Making (MCDM). One of the major obstacles in conducting MCDM studies is accessing subject domain experts, a challenge that can be mitigated by substituting traditional experts with virtual experts offered by LLMs such as ChatGPT. This substitution not only alleviates the scarcity of experts but also enhances the scalability and speed of the decision-making process. Moreover, GenAI can facilitate a more dynamic and adaptable MCDM environment by rapidly processing and synthesizing large datasets to evaluate alternatives under multiple criteria. This capability enables more informed and robust decision-making across industries, paving the way for smarter, data-driven solutions that reflect a true integration of Artificial Intelligence (AI) with human-like reasoning (5).

This study aims to showcase this potential by integrating AHP (Analytic Hierarchy Process) and ChatGPT. In this work, AHP will be applied to select suppliers for a machinery manufacturing setting. To complete this, the AHP will be conducted in the traditional way, where experts rate their criteria, and their results are compared with virtual experts developed by ChatGPT. Therefore, this paper has the following contributions:

- Introduction of the AIDM (AI-Driven Decision Making) framework, which utilizes GenAI for multiple MCDM analysis tasks.
- Demonstration of AIDM's integration of AHP and ChatGPT, showcasing its effectiveness in supplier selection.

2 Literature Review

The satisfactory performance of LLMs has inspired MCDM researchers to introduce advanced AI capabilities into traditional decision frameworks. As a result, there is a growing body of literature exploring the application

of LLMs like ChatGPT in automating and refining decision processes that traditionally required extensive human judgment.

Pandey et al. (2024) explore the integration of intuitionistic fuzzy DEMATEL with ChatGPT to analyze AI's impact on automation, focusing on how AI can enhance decision-making systems by capturing complex interrelationships between criteria (1). This study demonstrates how generative AI can provide a nuanced understanding of decision dynamics in highly automated environments. Wang and Wu (2024) present an innovative approach where ChatGPT serves as a virtual expert in the multi-criteria decision-making process for supplier evaluation (3). Their work highlights ChatGPT's ability to analyze extensive criteria, offering a reliable alternative to human expert judgments in industrial settings.

Svoboda and Lande (2024) enhance MCDM by integrating the AHP with GPT-4, focusing on automating decision support to improve efficiency and effectiveness across various sectors (6). Zuheros et al. (2024) examine LLMs for crowd decision-making, utilizing ChatGPT to facilitate collaborative decision processes (5). They provide a framework that leverages the collective intelligence of crowds, enhanced by AI's processing capabilities, to improve decision accuracy and inclusivity. Chuma and de Oliveira (2023) investigate generative AI's role in business decision-making, with a specific focus on how ChatGPT can streamline strategic planning and operational decisions within corporations (4). Their study illustrates the practical applications of AI in enhancing business efficiency through improved decision-making speed and quality.

These diverse applications of LLMs in decision-making illustrate the broad potential and adaptability of AI technologies across different domains and decision-making frameworks. Collectively, these studies provide strong evidence of the benefits and future possibilities for AI in enhancing traditional decision processes.

3 AI-driven Decision-Making (AIDM) Framework

Building on the advancements in AI for decision-making, this paper introduces a new framework that synergizes methodologies from recent studies, specifically those by Wang and Wu (3) and Svoboda and Lande (6). This proposed AIDM framework is designed to enhance decision accuracy and efficiency across various complex decision scenarios and can be adapted for different applications such as supplier selection, strategic business evaluations, or any other context requiring MCDM. The versatility of this framework allows for the integration of any MCDM model with GenAI technologies, providing a robust tool for complex decision-making processes in a diverse range of industries.

The following subsections delve into the core components of the AIDM framework, detailing the intricacies of its application in simulating and augmenting human expertise in decision-making.

3.1 Expert Identification

As depicted in Figure 1, the AI-driven Decision Making (AIDM) process harnesses a Language Model to virtually replicate the expertise traditionally provided by human experts in the MCDM approach. As part of the AIDM framework, the Expert Identification phase is crucial. This involves:

- **Prompt Development**: Crafting the prompts to establish the decision-making context for the AI.
  - *Context Specification*: Setting up the AI's environment by clearly outlining the problem space.
  - *Persona Crafting*: Creating a detailed virtual expert profile for the AI, enabling it to mimic human expert behavior.
  - *Instruction Configuration*: Providing the AI with precise instructions to guide its decision-making process.
- **Refinement Process**: Enhancing the AI's performance through iterative development.
  - *Testing*: Assessing the AI's decision-making capabilities through simulated scenarios.
  - *Iterative Feedback*: Using the insights gained from testing to refine the AI's decision-making processes.

3.2 Criteria Development

The Criteria Development phase is essential in the AI-Enabled Decision Making framework. It involves:

- **Objective Definition**: Setting clear goals for the AI to address in the decision-making process.
- **Criterion Identification**: Utilizing AI to simulate virtual experts who will identify key criteria.
- **Uniqueness Verification**: Filtering out duplicate criteria to ensure a unique set for evaluation.

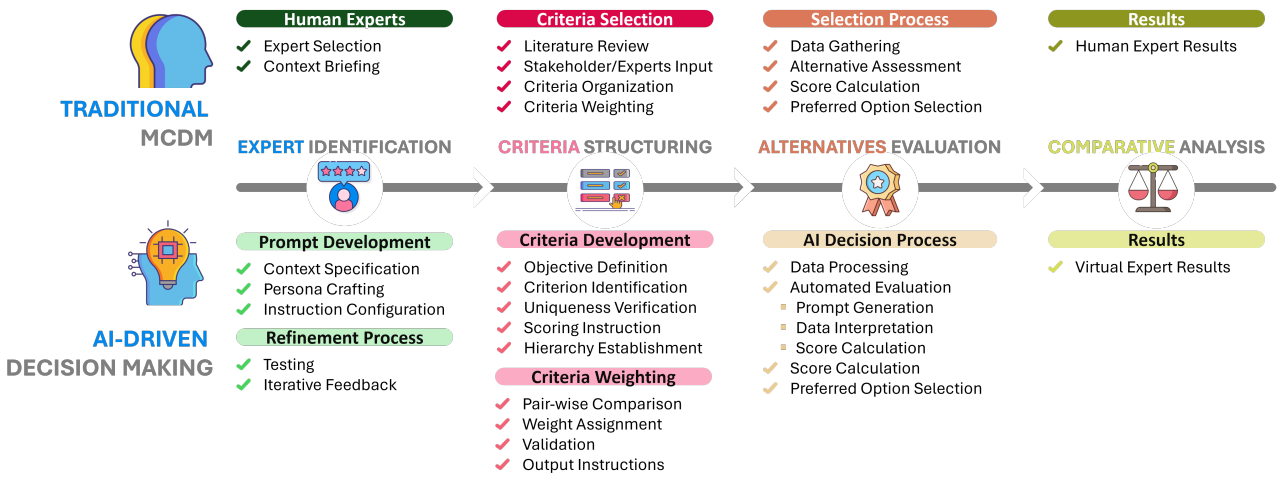

Figure 1: Detailed comparison of the procedural steps in Traditional MCDM versus the AI-driven Decision-Making (AIDM) framework, highlighting the transition from human expert reliance to AI-driven virtual experts in decision support systems.

- **Scoring Instruction**: Instructing virtual experts to score criteria using a scale where 1 indicates the least and 9 the most importance.
- **Hierarchy Establishment**: Organizing criteria into a hierarchical structure based on their relative significance.

Following the development of criteria, the Criteria Weighting phase is conducted. This phase encompasses:

- **Pair-wise Comparison**: AI, through virtual experts, conducts pair-wise comparisons of criteria based on their importance.
- **Weight Assignment**: The AI system assigns weights to each criterion based on the outcome of the comparisons.
- **Validation**: Validation processes are implemented to ensure the consistency and reliability of the assigned weights.
- **Output Instructions**: Detailed instructions are provided to the AI system to output the findings in a structured format, often an Excel file, for further analysis and verification.

3.3 Alternatives Evaluation

The last subsection is Alternatives Evaluation, which encompasses the AI Decision Process. This phase involves:

- **Data Processing**: Utilize AI to interpret and analyze the information provided in response to the prompts. This involves understanding the context, extracting relevant data, and synthesizing it into a structured format for further analysis.
- **Automated Evaluation**:
  - *Prompt Generation*: Craft prompts to establish the decision-making context for the AI.
  - *Data Interpretation*: Use AI to interpret and analyze the data gathered from the prompts.
  - *Score Calculation*: Calculate scores for each alternative based on the evaluated data.
- **Preferred Option Selection**: Select the preferred option based on the calculated scores and decision criteria.

3.4 Comparative Analysis of Decision-Making Results

In the final phase of the AIDM framework, engagement in a **Comparative Analysis** contrasts the outcomes derived from AI-driven virtual experts with those obtained from human experts. This critical evaluation benchmarks the effectiveness of the AI model and provides insights into the consistency and reliability of the virtual decision-making process, ensuring that the AI's performance aligns with human expert standards.

The integration of ChatGPT and similar GenAI technologies within the AIDM framework marks a significant innovation in the field of decision-making. By harnessing the advanced capabilities of AI, this framework introduces a new paradigm that significantly improves decision accuracy and efficiency. The versatility of AI-driven processes allows for rapid adaptation to diverse decision scenarios, providing a substantial benefit over traditional methods. The iterative and dynamic nature of AIDM also ensures continuous learning and improvement, paving the way for increasingly sophisticated decision support systems.

Moreover, the AIDM framework's flexibility enables it to encompass any MCDM methodology, such as the Analytic Hierarchy Process (AHP), Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), Decision Making Trial and Evaluation Laboratory (DEMATEL), among others. GenAI technologies like ChatGPT are adept at emulating these complex decision processes. Advanced versions, such as ChatGPT-4, can analyze original academic or foundational texts, assimilating the nuances of these methodologies to refine the decision-making models. This allows for a nuanced application that aligns closely with the intricate nature of established decision-support frameworks.

4 Case Study: AIDM in Supplier Selection

The case study on supplier selection in a heavy machinery manufacturing setting illustrates the AIDM framework's applicability. Recognized within the MCDM domain, supplier selection has been explored using different MCDM approaches, including AHP (7), TOPSIS (8), and DEMATEL (9). The AHP method, one of the most applied MCDM models, introduced by Saaty 1980. (10), employs a pairwise comparison method designed to elicit consistent relative judgments (11).

4.1 Expert Identification

As depicted in the AIDM framework, the first step is to identify experts. The crucial part of this phase is writing a well-defined prompt so that Chat-GPT can provide accurate information based on the problem context. As outlined in Figure 2, which illustrates the prompt development necessary for generating a team of virtual AI agents. These agents, named Expert 1 to Expert 5, are designed to have diverse backgrounds to cover all aspects of supplier selection in the heavy machinery manufacturing sector. The prompt ensures a comprehensive understanding and unique insight from each virtual expert.

Prompt Development

This prompt aims to facilitate the creation of virtual AI agents tasked with streamlining supplier selection processes within the heavy machinery manufacturing industry.

- ✔ **Context Specification:** The AI agents will assist in identifying criteria and suppliers that can provide high-quality and durable ***<hydraulic systems>***. The agents should prioritize suppliers with robust supply chains and a history of timely deliveries.
- ✔ **Persona Crafting:** We aim to create virtual experts embodying the expertise of supply chain managers with extensive knowledge in the ***<manufacturing sector>***. These experts are adept at evaluating suppliers based on some factors that will be defined in the next step.
- ✔ **Instruction Configuration:** Create ***<five>*** different agents, each with a slightly different background to offer unique insights during the evaluation process. Name these experts ***<Expert 1-5>***. These experts should cover diverse perspectives. List these experts in a **<table format>**.

Identified AI Agents by ChatGPT-4

Expert	Background	Area of Expertise	Unique Insight
Expert 1	Extensive experience in hydraulic engineering	Technical assessment of hydraulic systems	Offers in-depth technical analysis of products and materials used
Expert 2	Background in supply chain logistics	Supply chain reliability and efficiency	Focuses on the robustness and resilience of the supplier's supply chain
Expert 3	Worked in quality control for heavy machinery	Quality assurance and control standards	Prioritizes suppliers with the highest quality certifications and reviews
Expert 4	Experience with international suppliers	Global market trends and international sourcing	Brings insights on global supplier networks and risk management
Expert 5	Specialization in sustainable manufacturing	Eco-friendly and sustainable supply practices	Evaluates suppliers based on their sustainability and environmental impact

Refinement Process

In case the expert results are not satisfactory, the prompt can be refined and adjusted based on the feedback received. Modifications will be made iteratively to the prompt to ensure satisfactory outcomes.

Figure 2: Prompt example for developing virtual AI experts, showcasing the information profile provided by ChatGPT-4. The highlighted parts of the prompt, indicated by <...>, are some of the adjustable inputs that can be changed during the refinement process to achieve the desired combination of virtual experts. virtual experts.

4.2 Criteria Development

Once the virtual AI experts are identified, ChatGPT can be applied to provide a list of important criteria for the selected problem, namely supplier selection. Leveraging the capabilities of ChatGPT-4 and its API, all of the Criteria Development tasks are completed within a single prompt. The prompt comprehensively guides the AI through the process, ensuring that all relevant aspects are covered. As depicted in Figure 3, this prompt facilitates the creation of a structured list of criteria that are essential for evaluating potential suppliers.

Criteria Development Prompt

The objective of this prompt is to develop criteria for the supplier selection process using ChatGPT. The criteria will be generated based on the expertise of virtual AI experts, named Expert 1 through Expert 5.

- ✔ **Criterion Identification:** For each of the developed virtual experts (Expert 1, ..., Expert 5), criteria for supplier selection will be developed based on their backgrounds. **<8>** criteria will be developed for each expert. The results will be saved in an Excel worksheet called "Criterion Identification" with two columns: Expert name and the criteria.
- ✔ **Uniqueness Verification:** Duplicate criteria will be removed, and criteria with similar meanings will be merged. Unique criteria will be listed **<max 20>**, and the results will be saved into a new tab of Excel called "Unique Criteria".
- ✔ **Scoring Instruction:** Based on the unique criteria results, all virtual experts will rate the importance of each criterion from 1 (least important) to 9 (most important). The results of scoring will be tabulated in a new worksheet, with a column for criteria and a column for each expert (Expert 1, ..., Expert 5). Total scores for each criterion will be calculated as the sum of all scores for that criterion based on experts' ratings. The last column will include the total score for each criterion. Criteria will be sorted based on their total score from largest to smallest and saved in a new worksheet.
- ✔ **Hierarchy Establishment:** The top **<15>** high scored criteria will be considered and categorized into three categories to create the hierarchy. They will be listed in the Excel file. All worksheets will be part of the same Excel file.

Figure 3: Criteria development process utilizing ChatGPT-4.

The hierarchy of supplier selection criteria is a crucial aspect of the decision-making process. As illustrated in Figure 4, the criteria have been categorized into three primary dimensions: Quality, Cost, and Attributes. Each of these dimensions is further broken down into specific factors that contribute to the overall assessment of potential suppliers. The Quality dimension encompasses criteria such as the inherent quality of goods or services, reliability, communication, production flexibility, and customer feedback. The Cost dimension includes direct cost, lead time, frequency of pricing updates, and contract terms. Lastly, the Attributes dimension captures environmental impact, accessibility, origin of the supplies, references provided, management systems in place, and the global expertise of the supplier. This structured approach aids in systematically evaluating and comparing suppliers to select the most suitable one based on a comprehensive set of criteria.

The subsequent step is to perform the pairwise comparison and assign weights, a meticulous task when carried out manually, as it requires each expert to evaluate and score criteria against each other. Additionally, considerations such as the Consistency Ratio (CR) must be taken into account to ensure the validity of the assessments.

With the advent of Generative AI, this process can be significantly optimized. As illustrated in Figure 5, a well-crafted prompt with ChatGPT-4 can perform all tasks associated with the Analytic Hierarchy Process (AHP), including conducting the analysis and performing the calculations necessary to obtain the weights. Notably, since ChatGPT-4 is employed, the platform has access to a wealth of online resources and can understand

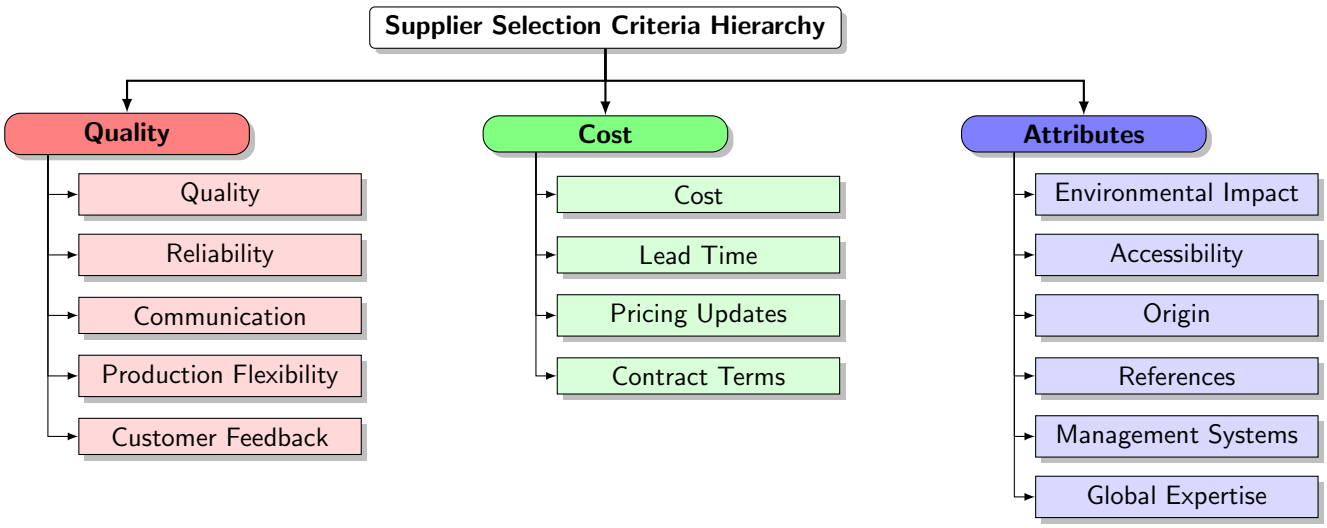

Figure 4: Supplier Selection Criteria Hierarchy.

the procedure of the calculation. It can process inputs such as a diagram of criteria hierarchy or the table of expert evaluations to provide weights corresponding to the defined problem.

This approach does not only complete the step-by-step calculation but also generates a detailed Excel file that includes all the details required for verification and validation. Should the results not meet the expected standards, the prompt can be adjusted until satisfactory results are achieved. This makes the calculation process not only efficient but also adaptable, ensuring the methodology aligns with the nuanced requirements of the decision-making scenario.

Criteria Weighting Prompt

Conduct a revised Analytic Hierarchy Process (AHP) developed by Saaty in 1980 for supplier selection based on updated criteria and expert inputs. The goal is to assess suppliers for heavy machinery manufacturing by comparing multiple criteria across quality, cost, and other attributes. Assume that I will provide the list of criteria and the background information for each expert.

- **Pair-wise Comparison:** Each expert will compare the criteria within their specific dimension of expertise using Saaty's scale. The comparisons will be between pairs of criteria related to Quality, Cost, and other Attributes. I will provide you with the list of criteria and the experts' specific areas of expertise.
- **Weight Assignment:** Assign weights to the pairs of criteria based on the experts' comparisons. Utilize Saaty's scale where 1 represents equal importance and 9 represents extreme importance, with intermediate values for nuanced preferences. The weights should reflect the experts' assessments based on their backgrounds.
- **Validation:** Ensure the consistency of the pairwise comparisons and weights. Validate that the weights are reasonable and align with the expertise of the relevant expert. If inconsistencies are detected, indicate the need for reassessment.
- **Output Instructions:** Generate an Excel worksheet that includes the following sheets:
  1) Pairwise Comparisons: For each expert's scores.
  2) Aggregated Comparison Matrix: Combining scores from all experts.
  3) Normalized Comparison Matrix: To obtain comparable weights.
  4) Criteria Weights: Calculating the final weights of each criterion.
  5) Provide the file in a downloadable format and advise on any inconsistencies found during validation.

Figure 5: The Analytic Hierarchy Process facilitated by ChatGPT-4, showcasing the streamlined calculation and validation process.

As part of the Criteria Weighting phase, a snapshot of the criteria pairwise comparison provided by the Generative Pre-trained Transformer 4 is depicted in Figure 6. The table showcases the varied perspectives of different virtual experts on the importance of each criterion relative to others within the 'Quality' dimension. This exemplifies the use of the platform to simulate a comprehensive expert panel evaluation, streamlining the Analytic Hierarchy Process.

	A	B	C	D	E	F	G	H	I
1	**Dimension**	**Criterion 1**	**Criterion 2**	**Expert 1**	**Expert 2**	**Expert 3**	**Expert 4**	**Expert 5**	**Geometric Average**
2	Quality	Quality	Reliability	1.40	1.00	1.29	1.00	1.00	1.12
3	Quality	Quality	Communication	7.00	0.33	9.00	1.00	1.00	1.84
4	Quality	Quality	Production Flexibility	7.00	1.00	9.00	1.00	1.00	2.29
5	Quality	Quality	Customer Feedback	7.00	1.00	3.00	1.00	1.00	1.84
6	Quality	Quality	Cost	2.33	0.14	9.00	1.00	1.00	1.25
7	Quality	Quality	Lead Time	7.00	0.20	9.00	1.00	1.00	1.66

Figure 6: A snapshot of the pairwise comparison for 'Quality' criteria as generated by ChatGPT.

4.3 Alternatives Evaluation

Once the criteria are determined and assigned weights, alternative evaluation can be performed. Regardless of the approach—whether through human expertise or virtual AI experts—the evaluation involves a systematic comparison of potential suppliers against the established criteria. This section, however, is omitted from this article since the primary aim of the paper is to demonstrate the application of AIDM for determining criteria and weights. The selection process itself can be conducted using any specific MCDM approach. The focus here is to highlight the efficiency and effectiveness of AIDM in the critical preparatory steps that precede the actual supplier selection, ensuring that the subsequent MCDM process is grounded on a robust and well-defined set of criteria.

4.4 Comparative Analysis

The evaluation of alternatives is followed by a comparative analysis, which serves to juxtapose the weights attributed to the selection criteria by AIDM, involving 5 virtual experts, against those assigned by human experts, comprising 8 members. This juxtaposition not only illuminates the contrast in decision-making but also

underscores the diverse perspectives that each group may bring to the process. Table 1 delineates the respective weights and the computed differences, thus providing a quantified look at the distinctions in prioritization between the two evaluative bodies.

As shown in Table 1, a comparative analysis of supplier selection criteria was conducted, delineating the percentage weights determined by AIDM (Automated Intelligent Decision Making) alongside those assessed by human experts in Domestic and International Purchasing contexts.

Table 1: Comparative weights of supplier selection criteria as determined by AIDM and human experts, with percentage point differences.

Criterion	1-Domestic Team (%)	2-Int Team (%)	3-AIDM (%)	Diff (1,2) (%)	Diff (1,3) (%)
Customer Feedback	2.5	5.0	6.0	−3.5	−1.0
Pricing Updates	9.5	10.0	4.8	4.7	5.2
References	2.5	5.0	6.0	−3.5	−1.0
Reliability	11.4	13.9	9.8	1.6	4.1
Quality	14.6	11.2	11.1	3.5	0.1
Environmental Impact	2.3	1.6	7.5	−5.2	−5.9
Origin	2.9	2.4	4.8	−1.9	−2.4
Lead Time	12.3	6.1	6.7	5.6	−0.6
Communication	5.4	9.3	6.0	−0.6	3.3
Production Flexibility	4.5	3.3	4.8	−0.3	−1.5
Accessibility	4.6	9.4	4.8	−0.2	4.6
Cost	14.0	7.4	8.9	5.1	−1.5
Global Expertise	2.0	3.0	7.1	−5.1	−4.1
Contract Terms	6.9	8.1	4.8	2.1	3.3
Management Systems	3.6	2.7	6.7	−3.1	−4.0

This table provides valuable insights into the relative importance assigned to various supplier selection criteria by both automated decision-making algorithms and human experts.

The table articulates notable discrepancies in criteria weights, such as those for 'Pricing Updates' and 'Environmental Impact,' which indicate a stark contrast in the emphasis placed by AIDM versus human experts. These differences may stem from the distinct methodologies inherent to the AIDM's algorithmic processing as opposed to the subjective judgement of human experts. The analysis underscores the importance of selecting the appropriate decision-making strategy that aligns with organizational goals and the nature of the decision context.

5 Conclusion

This paper has endeavored to introduce and exemplify the novel concept of AI-Enabled Decision Making, AIDM. Through the case study of supplier selection, the paper has demonstrated the practical implementation of AIDM and its potential to streamline and enhance the decision-making process. It is clear that AIDM offers a significant advantage in terms of analytical capability and efficiency.

However, the field is emerging, and there remains ample scope for future enhancements. One immediate improvement could be the development of a dedicated API that simplifies and automates this process, allowing for seamless integration of AIDM into existing business systems. Another avenue could be the creation of a more interactive platform where decision-makers can input real-time data and receive instantaneous recommendations from the AIDM system. Furthermore, expanding the AIDM framework to incorporate real-time data analytics would provide an even more dynamic and responsive decision-making tool.

As businesses continue to navigate an ever-complex landscape, the deployment of AIDM could be pivotal in securing a competitive edge. The advancements proposed herein aim not only to augment the effectiveness of decision-making but also to equip leaders with a more robust and intelligent toolkit to address future challenges.

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